Retrieval-Augmented Input-Space Translation for Frozen Clinical Predictors

NeurIPS 2026 — Positioning Document

Abstract

Clinical prediction models validated at one hospital degrade when deployed at another due to distributional shift in electronic health records (EHR). Standard domain adaptation methods require modifying the target model—incompatible with regulatory frameworks that mandate locked clinical algorithms. We propose *input-space translation* for frozen clinical predictors: learning a neural translator that maps source EHR time series into the target input distribution while the predictor remains *entirely untouched*. We develop three architectures of increasing capability—delta-based residual, shared latent encoding, and retrieval-augmented cross-attention grounded in a target-domain memory bank—and introduce the *gradient alignment condition* $\alpha(\theta) = \cos(\nabla\mathcal{L}_{\text{task}}, \nabla\mathcal{L}_{\text{fid}})$, a diagnostic measurable in the first training epoch that predicts when frozen-model translation will succeed or fail. Across five clinical tasks on eICU→MIMIC-IV, the retrieval translator achieves universal improvements of up to $+5.6$ AUROC and $+16.1$ AUCPR over the frozen baseline: on AKI, it surpasses even the *in-domain* MIMIC-native LSTM in both AUROC (91.14 vs. 89.7) and AUCPR (72.9 vs. 66.5), all three classification tasks exceed the eICU-native LSTM on both metrics, and length-of-stay regression matches the eICU-native model—all with zero modification to the frozen predictor. To the best of our knowledge, this is the first work to achieve cross-hospital domain adaptation for clinical time series while keeping the target predictor entirely frozen, enabled by instance-level retrieval from the target domain and a gradient alignment theory that predicts when frozen-model translation succeeds.

1 Related Work and Current Methods

Domain adaptation in clinical and general time series. Domain adaptation for electronic health records has been approached through representation-space alignment Purushotham et al. [2017], Zhang et al. [2022], ExtraCare Authors [2026], optimal transport OTTEHR Authors [2025], inductive transfer with model fine-tuning Mutnuri et al. [2024], Driever et al. [2025], and data harmonization pipelines Purushotham et al. [2018], Harutyunyan et al. [2019]. VRADA Purushotham et al. [2017] introduced variational RNNs with adversarial alignment for clinical time series but requires end-to-end training of the target model. OTTEHR OTTEHR Authors [2025] applies optimal transport to the same eICU–MIMIC databases we study, yet performs population-level feature alignment without temporal modeling and without any frozen-model constraint. ExtraCare ExtraCare Authors [2026], a concurrent work on interpretable clinical DA through concept-grounded orthogonal decomposition, also operates in representation space. In the broader time-series DA literature, established methods—CoDATS Wilson et al. [2020], RAINCOAT He et al. [2023], CLUDA Ozyurt et al. [2023], ACON Liu et al. [2024a]—learn domain-invariant representations through a shared feature extractor, typically with adversarial Wilson et al. [2020], contrastive Ozyurt et al. [2023], or spectral kernel Liu and Xue [2021] objectives. ADATIME Ragab et al. [2023a] benchmarks eleven such methods across fifty scenarios, all operating in representation space. MAPU Ragab et al. [2023b] addresses the orthogonal constraint of restricted *data* access (source-free adaptation) but still modifies the target model. A unifying limitation of every method above is the assumption that the target model's parameters are trainable: they either fine-tune the predictor or jointly learn a feature extractor. When the target predictor must remain frozen—as required for regulatory compliance with locked clinical algorithms under the FDA's Software as a Medical Device (SaMD) framework U.S. Food and Drug Administration [2021a]—these methods are inapplicable, motivating input-space adaptation approaches.

Frozen-model adaptation and input-space methods. A growing body of work modifies *inputs* rather than model weights. Prompt tuning Lester et al. [2021] learns a fixed soft-token prefix for frozen language models; Visual Prompt Tuning (VPT) Jia et al. [2022] extends this to frozen vision transformers. Adversarial reprogramming Elsayed et al. [2019] repurposes frozen ImageNet classifiers via a single universal additive perturbation. Voice2Series Yang et al. [2021] reprograms frozen acoustic models for time-series classification, and Time-LLM Jin et al. [2024] reprograms frozen LLMs (GPT-2, LLaMA) for time-series forecasting—the most prominent frozen-model work in the time-series literature, with over 1000 citations. TATO Qiu et al. [2026], a concurrent ICLR 2026 contribution, explicitly proposes "adapt data to model" for frozen time-series foundation models using handcrafted transformation operators (slicing, normalization, outlier correction). L2C Chi et al. [2025] learns input-space modifications for frozen CLIP at test time. PixelDA Bousmalis et al. [2017] performs input-space DA in vision via GANs but does not freeze the downstream model. SHOT Liang et al. [2020] partially freezes the classifier head while adapting the feature extractor; our constraint is stronger in that the *entire* model is frozen. Table 1 summarizes the key distinctions. These methods face three limitations in our setting. First, prompt tuning, VPT, and reprogramming apply *input-independent* transformations: they learn a single perturbation that is shared identically across all inputs. While our translator also has fixed (learned) parameters, its output is *input-conditioned*—each patient's time series receives a distinct transformation adapted to its specific values, length (1 to 169 timesteps), and missingness pattern across 48+ features. A universal additive perturbation cannot provide such sample-level adaptation, as the correction needed for one patient's lab values may be entirely inappropriate for another's. Second, the vast majority target *task* adaptation (repurposing a model for a new task), not *domain* adaptation (translating data from one hospital's distribution to another's). Third, handcrafted transforms (TATO) rely on a predefined set of

Table 1: Comparison of frozen-model and input-space adaptation methods. Our approach is the only one that combines sample-conditional neural translation, target-domain retrieval, variable-length support, and cross-domain DA for clinical time series.

Method	Input modification	Sample-cond.	Retrieval	Var.-length	Purpose
Prompt Tuning Lester et al. [2021]	Fixed prefix	✗	✗	✗	Task adapt.
VPT Jia et al. [2022]	Prepended tokens	✗	✗	✗	Task adapt.
Voice2Series Yang et al. [2021]	Learned transform	✗	✗	✗	Cross-task
Time-LLM Jin et al. [2024]	Reprogramming layer	✗	✗	Fixed patches	Cross-modality
TATO Qiu et al. [2026]	Handcrafted pipeline	Partially	✗	✓	Domain adapt.
Ours	**Neural translator**	✓	✓	✓	**Cross-domain DA**

operators (slicing, normalization, outlier correction) applied independently per feature. Clinical domain shift, however, involves nonlinear, *context-dependent* distributional differences: the correction required for a lab value depends on concurrent vitals, medications, and the patient's trajectory—interactions that a fixed pipeline of feature-wise operators cannot model. To our knowledge, no prior frozen-model work simultaneously incorporates instance-level retrieval from the target domain, evaluates on clinical EHR data, and provides theoretical grounding via gradient dynamics. These capacity limitations motivate our neural translator, which uses instance-level retrieval from the target domain for contextual grounding.

Retrieval-augmented models for time series. Retrieval-augmented generation (RAG) Lewis et al. [2020] established the paradigm of conditioning neural models on retrieved external knowledge. kNN-LM Khandelwal et al. [2020] augments a frozen language model with nearest-neighbor retrieval from a token-level datastore, achieving domain adaptation by simply swapping the datastore—without any model retraining. kNN-MT Khandelwal et al. [2021] extends this to machine translation, performing per-timestep k-NN lookup from a domain-specific datastore and interpolating *output token distributions* with the frozen model's predictions. RETRO Borgeaud et al. [2022] introduced chunked cross-attention over retrieved document passages—the closest architectural precedent for our fusion mechanism. However, RETRO retrieves passages from the *same* domain to augment language generation, whereas we retrieve from the *target* domain to transform *source* inputs, making retrieval an instrument of domain translation rather than knowledge augmentation. In the time-series literature, retrieval augmentation is an emerging paradigm at top venues: RATD Liu et al. [2024b] retrieves relevant series to guide diffusion-based forecasting (NeurIPS 2024), TS-RAG Ning et al. [2025] augments time-series foundation models with an adaptive retrieval mixer (NeurIPS 2025), and RAFT RAFT Authors [2025] exploits inductive biases from retrieved examples for forecasting (ICML 2025). RAM-EHR Xu et al. [2024] retrieves textual medical knowledge from PubMed and knowledge graphs to augment EHR predictions, but retrieves *text*, not time-series data. Our retrieval translator differs from all prior work in three respects: (a) we retrieve encoded *target-domain time-series windows*—representations of what the frozen predictor expects to see—not text passages or output distributions; (b) retrieved windows serve as *cross-attention context* in a Transformer decoder, enabling selective feature-level borrowing, rather than being interpolated with output distributions (kNN-MT) or concatenated with the query (RAG); and (c) the purpose is cross-domain translation, not knowledge augmentation or forecasting. kNN-MT Khandelwal et al. [2021] is the closest ancestor: both systems pair a frozen model with per-step k-NN from a domain-specific store. The critical difference is that kNN-MT interpolates output token distributions at the decoder, operating within a single language, while our retrieval translator uses retrieved embeddings as cross-attention context to transform the *input*, bridging fundamentally different data distributions (eICU vs. MIMIC-IV feature spaces with heterogeneous recording practices, device calibrations, and documentation conventions).

Foundation models as an alternative paradigm. Large-scale pre-training offers a complementary path to cross-hospital generalization. ICareFM Burger et al. [2025] pre-trains on 650K ICU stays from multiple hospitals with a self-supervised time-to-event objective, achieving zero-shot transfer. CLMBR/EHRSHOT Wornow et al. [2023] provides a 141M-parameter EHR foundation model with strong few-shot evaluation on 6,739 patients. LLEMR Wu et al. [2024] instruction-tunes an LLM on 400K+ MIMIC-IV examples. Med-BERT Rasmy et al. [2021] and BEHRT Li et al. [2020] pre-train Transformers on 28.5M and 1.6M patient records, respectively. We build directly on the YAIB benchmark framework van de Water et al. [2024], whose multi-center preprocessing pipeline and five standardized task definitions provide our experimental infrastructure and frozen baselines (LSTMs in our experiments, though the input-space translation paradigm is architecture-agnostic). Foundation models require (a) massive multi-center datasets (100K–650K stays), (b) weeks of compute for pre-training, and (c) replacing the deployed predictor entirely, necessitating full regulatory re-validation. Our approach is complementary, not competitive: when a hospital has a validated, deployed predictor that cannot be retrained due to regulatory, institutional, or practical constraints, input-space translation is the only viable adaptation strategy. Foundation models and our translators address different points on the *deployment constraint spectrum*—from unconstrained model replacement to fully frozen predictor preservation.

Domain adaptation theory and gradient dynamics. Ben-David et al. Ben-David et al. [2010] bound target-domain error by $\varepsilon_T \leq \varepsilon_S + d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{S}, \mathcal{T}) + \lambda^*$, where $d_{\mathcal{H}\Delta\mathcal{H}}$ measures distribution divergence and λ^* is the ideal joint error. Mansour et al. Mansour et al. [2009] refine this via discrepancy distance tailored to the loss and hypothesis class; Zhang et al. Zhang et al. [2019] bridge theory and algorithms through margin disparity discrepancy. Canonical methods operationalize these bounds: DANN Ganin et al. [2016] minimizes $\mathcal{H}$-divergence via gradient reversal, Deep CORAL Sun and Saenko [2016] aligns second-order feature statistics, and DAN Long et al. [2015] matches multi-kernel MMD Gretton et al. [2012b] across layers. All require a *learnable* feature extractor. In our frozen-model setting, representation-space alignment is impossible by construction: since $\nabla_\phi f_T = 0$, no gradient signal can reshape the frozen model's internal representations, and any alignment objective (MMD, adversarial, CORAL) applied to intermediate layers has no learnable pathway to reduce the domain gap. We instead adapt these

methods to operate in input space and evaluate them as baselines (detailed in the full paper). The multi-objective structure of our training—minimizing task loss $\mathcal{L}_{\text{task}}$ (prediction accuracy through the frozen model) and fidelity loss $\mathcal{L}_{\text{fid}}$ (proximity to the source distribution) simultaneously—connects to the gradient conflict literature in multi-task learning. Sener and Koltun Sener and Koltun [2018] frame this as multi-objective optimization; PCGrad Yu et al. [2020] defines conflict as $\cos(\nabla\mathcal{L}_i, \nabla\mathcal{L}_j) < 0$ and projects conflicting gradients onto the normal plane; CAGrad Liu et al. [2021] provides convergence guarantees under conflict. In the DA literature specifically, Du et al. Du et al. [2024] analyze obtuse-angle gradients between alignment and classification losses in standard UDA and propose gradient rotation (GH/GH++), while Phan et al. Phan et al. [2024] align per-domain gradients for prompt-based adaptation of frozen vision-language models. Our contribution extends gradient conflict analysis to the frozen-model DA setting. We show that the cosine similarity $\cos(\nabla_\theta\mathcal{L}_{\text{task}}, \nabla_\theta\mathcal{L}_{\text{fid}})$ between task and fidelity gradients—measured at the translator parameters θ—is a diagnostic that predicts method–task suitability: $+0.84$ for mortality (aligned; delta translation suffices) versus -0.21 for sepsis (conflicting; delta translation fails catastrophically). Crucially, where PCGrad and CAGrad intervene *algorithmically* by manipulating gradient directions, our retrieval translator resolves the conflict *architecturally*: the cross-attention pathway over target-domain windows provides an alternative information channel that reduces the dependence of the task gradient on input-space fidelity, substantially mitigating the destructive interference without requiring gradient surgery (detailed empirical gradient measurements in the full paper).

To summarize, no prior work combines a frozen target predictor, learned neural input-space translation, instance-level target-domain retrieval, and gradient alignment theory for clinical time-series domain adaptation. The frozen-model constraint is motivated by real-world regulatory requirements, yet it renders all existing clinical DA methods, general time-series DA methods, and foundation model approaches inapplicable without fundamental modification. Our retrieval-augmented translator fills this gap: it operates entirely in input space, grounds translations in target-domain examples, and provides a theoretical framework that explains when and why the approach succeeds.

2 Problem Formulation

Frozen-Model Translation Setting. Let $\mathcal{X}_S \sim P_S$ denote source-domain clinical time series (eICU) and $\mathcal{X}_T \sim P_T$ denote the target domain (MIMIC-IV). A target-domain predictor $f_T : \mathbb{R}^{T\times F} \to \mathbb{R}^{T\times C}$ has been trained on $\mathcal{X}_T$ and is *entirely frozen*: all parameters satisfy $\nabla_\phi f_T = 0$ and are never updated. We learn a translator $g_\theta : \mathbb{R}^{T\times F} \to \mathbb{R}^{T\times F}$ that maps source inputs into the target input space so that the composite system $f_T \circ g_\theta$ performs well on source labels y_S. The composite training objective is:

$$\mathcal{L}(\theta) \;=\; \mathcal{L}_{\text{task}}\big(f_T(g_\theta(x_S)),\, y_S\big) \;+\; \lambda_{\text{fid}}\, \mathcal{L}_{\text{fid}}\big(g_\theta(x_S),\, x_S\big) \;+\; \lambda_{\text{range}}\, \mathcal{L}_{\text{range}}\big(g_\theta(x_S)\big), \tag{1}$$

where $\mathcal{L}_{\text{task}}$ is cross-entropy (classification) or MSE (regression) evaluated through the frozen predictor, $\mathcal{L}_{\text{fid}}$ is an MSE fidelity penalty that regularises the translator toward identity, and $\mathcal{L}_{\text{range}}$ penalises outputs outside the physiological range observed in $\mathcal{X}_T$.

The frozen constraint is motivated by regulatory and deployment considerations. Under the FDA Software-as-a-Medical-Device (SaMD) "locked algorithm" pathway, the downstream predictor is a validated, auditable artifact whose parameters must not change post-deployment U.S. Food and Drug Administration [2021b]. By learning only the input mapping g_θ, clinical sites can adapt a certified predictor to local data distributions without re-validation of f_T. This setting is strictly more constrained than test-time adaptation methods that update batch-normalisation statistics Wang et al. [2021] (which modify running mean and variance, altering the predictor's output for identical inputs) or partial model parameters Liang et al. [2020] (which retrain the feature extractor while freezing only the classifier head), and distinct from adapter-based approaches Houlsby et al. [2019] that insert learnable modules into the predictor's computation graph. In our setting, the predictor is treated as a fixed, deterministic function—no internal parameter, statistic, or architectural element is modified.

Gradient Alignment Condition. A central difficulty of frozen-model translation is that the task gradient must flow *through* the frozen predictor before reaching θ. Decomposing the gradient of Eq. 1:

$$\nabla_\theta\mathcal{L} \;=\; \nabla_\theta\mathcal{L}_{\text{task}} \;+\; \lambda_{\text{fid}}\, \nabla_\theta\mathcal{L}_{\text{fid}}, \tag{2}$$

we define the *gradient alignment coefficient*:

$$\alpha(\theta) \;=\; \cos\big(\nabla_\theta\mathcal{L}_{\text{task}},\; \nabla_\theta\mathcal{L}_{\text{fid}}\big). \tag{3}$$

Proposition 1 (Gradient Alignment Condition, informal)**.** *Stable frozen-model translation requires $\mathbb{E}[\alpha(\theta)] \geq 0$ during training. When $\alpha < 0$, the fidelity and task gradients exhibit destructive interference: reducing $\mathcal{L}_{\text{fid}}$ (moving the translation closer to identity) increases $\mathcal{L}_{\text{task}}$, while reducing $\mathcal{L}_{\text{task}}$ (improving predictions) increases domain divergence—violating the $\mathcal{H}\Delta\mathcal{H}$-divergence bound of Ben-David et al. [2010].*

We verify this empirically across five clinical tasks. Table 2 reports gradient diagnostics measured at the translator parameters during the first epoch of training.[1]

Across the three classification tasks, the alignment coefficient is the strongest predictor of translation success: mortality ($\alpha = +0.84$) and AKI ($\alpha > 0$) succeed because fidelity acts as a step-size regulariser along the task gradient direction, whereas sepsis ($\alpha = -0.21$) suffers destructive interference. Removing fidelity entirely ($\lambda_{\text{fid}} = 0$) causes catastrophic divergence (-0.101 AUROC, from 0.719 to 0.618), confirming that the task gradient alone is too noisy to guide the translator—the fidelity

[1] The full paper will extend these diagnostics across all three translator architectures and additional hyperparameter configurations, including per-epoch trajectories of $\alpha(\theta)$.

Table 2: Gradient alignment diagnostics across tasks. $\|\nabla_\theta \mathcal{L}_{\text{fid}}\|/\|\nabla_\theta \mathcal{L}_{\text{task}}\|$ is the fidelity-to-task gradient magnitude ratio; α is the cosine alignment (Eq. 3). Positive label rate is per-timestep for AKI/Sepsis and per-stay for Mortality.

Task	**Label rate**	$\|\nabla \mathcal{L}_{\text{fid}}\|/\|\nabla \mathcal{L}_{\text{task}}\|$	α	**Outcome**
Mortality	5.5% (per-stay)	2.8×	+0.84	+0.048 AUROC
AKI	11.95% (per-ts)	moderate	> 0	+0.056 AUROC
Sepsis	1.13% (per-ts)	5.7×	−0.21	+0.006 AUROC (delta)

term is necessary but, when misaligned, actively harmful. This tension motivates the progression from delta-based translation (Section 3) through shared-latent alignment (Section 3) to retrieval-guided translation (Section 3), where the retrieval mechanism provides an alternative information pathway that sidesteps the gradient alignment problem entirely.

3 Method

We present three translator architectures of increasing capability. Each addresses a specific limitation of its predecessor, producing a clear ablation path: delta (limited by gradient alignment) → shared latent (limited by reconstruction bottleneck) → retrieval-guided (resolves both).

4.1 Delta Translator. The delta translator produces additive corrections:

$$g_\theta^\delta(x) \;=\; x + \Delta_\theta(x), \tag{4}$$

where Δ_θ is parameterised by a stack of *AxialBlocks*—factored attention that alternates variable-wise attention (each timestep's F features attend to one another) and temporal attention (each feature's T timesteps attend across time). This factorisation reduces the attention cost from $\mathcal{O}((TF)^2)$ to $\mathcal{O}(T \cdot F^2 + F \cdot T^2)$ while preserving full expressivity Ho et al. [2019]. Static demographics (age, sex, height, weight) condition each block via FiLM layers Perez et al. [2018]: $h \mapsto \gamma \odot h + \beta$ with (γ, β) predicted from the static features. Temporal attention is *causal* (masked to prevent future information leakage) for per-timestep tasks (AKI, Sepsis, LoS) and *bidirectional* for per-stay tasks (Mortality, KidneyFunction).

The residual formulation ensures g_θ begins near identity—early training produces small perturbations, avoiding the divergence observed with $\lambda_{\text{fid}} = 0$. However, the delta translator is fundamentally limited by Proposition 1: when $\alpha < 0$ (sepsis), the task and fidelity gradients create destructive interference that caps performance at $+0.006$ AUROC despite extensive hyperparameter search (13 approaches $\times$ 2 tasks; detailed in the full paper).

4.2 Shared Latent Translator. To address gradient misalignment, the shared latent (SL) translator replaces the direct input-to-input mapping with an encoder–decoder architecture that passes through a shared bottleneck $z \in \mathbb{R}^{T \times d_{\text{latent}}}$. Training proceeds in two phases. **Phase 1** (autoencoder pretrain): the encoder–decoder is trained on MIMIC data only, with reconstruction and optional label prediction losses. **Phase 2** (translation): the full loss combines task, reconstruction, distributional alignment, and label prediction:

$$\mathcal{L}_{\text{SL}} = \mathcal{L}_{\text{task}} + \lambda_{\text{recon}}\, \mathcal{L}_{\text{recon}} + \lambda_{\text{align}}\, \text{MMD}(z_S, z_T) + \lambda_{\text{label}}\, \mathcal{L}_{\text{label_pred}}, \tag{5}$$

where MMD is the maximum mean discrepancy with a multi-scale Gaussian kernel Gretton et al. [2012a] and $\mathcal{L}_{\text{label_pred}}$ provides a direct label supervision signal that bypasses the frozen predictor. The MMD term provides distributional alignment at the latent level, replacing the input-space fidelity whose gradient opposes the task signal.

SL succeeds on tasks with sufficient label density: $+0.048$ AUROC ($+0.055$ AUCPR) on mortality (5.5% per-stay labels), $+0.054$ AUROC ($+0.133$ AUCPR) on AKI (11.95% per-timestep labels). However, it fails on sepsis (-0.017 to -0.043 AUROC), exposing a *reconstruction bottleneck*: the encoder–decoder must reconstruct all $T \times F$ features (e.g., 169×48) through the latent space. At sepsis's 1.13% positive rate, each training sequence contains $\sim$1 informative task gradient versus $\sim$8,000 reconstruction gradients—reconstruction dominates and the task signal cannot steer the latent representations. Crucially, this is not a per-timestep or causality issue: AKI is also per-timestep and causal, yet SL succeeds because its 11.95% label density provides $10\times$ more task gradient per sequence. This controlled comparison isolates *label density* as the root cause and motivates an architecture that does not rely on dense reconstruction to produce accurate translations.

4.3 Retrieval-Guided Translator. The retrieval-guided translator is our primary contribution. Instead of global distributional alignment (SL's MMD) or direct input perturbation (delta), it performs *instance-level retrieval*: each source timestep finds its nearest target-domain neighbours and selectively borrows feature-level information via cross-attention. This opens a different information pathway: the model need not "invent" correct target-domain feature values from noisy task gradients, but can instead attend to actual target-domain exemplars.

Memory Bank. All target-domain (MIMIC) training data is encoded through the shared encoder and segmented into temporal windows of size W (default W=6) with configurable stride s for overlapping coverage. Each window is mean-pooled to produce a compact descriptor:

$$M_j \;=\; \frac{1}{|W_j|} \sum_{t \in W_j} \text{Enc}(x_T^{(i)})_t \;\in\; \mathbb{R}^{d_{\text{latent}}}, \tag{6}$$

yielding a memory bank $M \in \mathbb{R}^{N_{\text{windows}} \times d_{\text{latent}}}$ that resides on GPU for fast retrieval. The memory bank is rebuilt every R epochs (default R=3), since the encoder evolves during training and a stale bank degrades retrieval quality. Overlapping windows (stride s=3, yielding $\sim$2$\times$ more windows) improve performance on AKI ($+0.003$ AUROC, $+0.010$ AUCPR), but excessive

overlap (stride s=1, ∼2.4M windows) degrades results (−0.002 AUROC), suggesting a coverage–redundancy tradeoff where moderate overlap improves recall of relevant exemplars but excessive redundancy dilutes the nearest-neighbour signal.

Per-Timestep k-NN Retrieval. For each source timestep t, we form a backward-looking query by mean-pooling source latents from $[\max(0, t-W+1),\ t+1)$, ensuring causal consistency. Learned importance weights $\sigma(\mathbf{w}) \in [0,1]^{d_{\text{latent}}}$ (sigmoid-gated, per latent dimension) modulate the distance metric:

$$d(q_t, M_j) \;=\; \left\| \sqrt{\sigma(\mathbf{w})} \odot (q_t - M_j) \right\|_2, \tag{7}$$

and the K nearest windows per timestep are retrieved (default K=16). Full timestep-level latents from each retrieved window are gathered to form the context tensor $C \in \mathbb{R}^{B\times T\times KW\times d_{\text{latent}}}$. Retrieval is per-timestep rather than per-sequence: we find that local temporal matching (window W=6) significantly outperforms wider windows (W=12: +0.046 vs. +0.056 AUROC on AKI), consistent with clinical time series where local physiological states are more informative than global trajectory similarity.

Cross-Attention Fusion. Retrieved context is integrated through L stacked cross-attention blocks. Each block performs three stages:

$$\begin{aligned} h'_t &= \text{CrossAttn}(Q{=}h_t,\ K{=}C_t,\ V{=}C_t) + h_t, \\ \tilde{h} &= \text{CausalSelfAttn}(h') + h', \\ \hat{h} &= \text{FFN}(\tilde{h}) + \tilde{h}, \end{aligned} \tag{8}$$

where h_t is the source latent at timestep t, C_t contains the KW retrieved target latents for that timestep, and all sub-layers use pre-LayerNorm residual connections. Stage 1 (per-timestep cross-attention) performs selective, feature-level borrowing from target exemplars—critically, this is *learned*, not interpolated. Compared to the kNN-MT approach of Khandelwal et al. [2021], which interpolates uniformly across all dimensions, cross-attention allows the model to attend selectively to different features from different neighbours. Stage 2 (global causal self-attention) enforces temporal coherence across the translated sequence.

The number of cross-attention blocks L is task-dependent and interacts with label density. On AKI (11.95% positive rate), increasing from L=2 to L=3 yields +0.009 AUROC—the additional capacity integrates richer target context from the dense supervision signal. On sepsis (1.13% positive rate), L=3 *hurts* by −0.006 AUROC: extra parameters overfit when gradient signal is sparse. This interaction provides a clean ablation of how architectural capacity must be matched to task supervision density.

Why Retrieval Resolves Gradient Alignment. The fundamental limitation of delta and SL translators is that the task gradient must specify both *which features* to correct and *what values* to produce—all by backpropagating through the frozen model's opaque hidden states. Retrieval decouples these two problems by introducing an orthogonal information channel. The memory bank supplies actual target-domain feature values; the model need only learn a soft *selection* over them via cross-attention weights. This constrains the output space to weighted combinations of real target exemplars, yielding a better-conditioned optimisation problem: the task gradient specifies *which* neighbours are useful (a selection problem over K candidates), while the neighbours themselves supply the *content* (correct feature values from the target distribution). Because the fidelity objective is now computed against outputs that are already grounded in the target domain, its gradient no longer opposes the task gradient—both objectives benefit from selecting more relevant target exemplars.

A critical design choice is detaching the source latent before querying the memory bank: $q_t = \text{Enc}(x_S)_t.\text{detach}()$. Since k-NN retrieval involves a non-differentiable argmin, gradients cannot propagate through the retrieval step; detaching prevents spurious gradient signals from the bank lookup. The encoder is still updated through the downstream cross-attention and decoding losses.

The empirical results confirm this analysis. On sepsis, where $\alpha = -0.21$ renders delta translation ineffective (+0.006 AUROC) and SL collapses (−0.017 to −0.043), retrieval achieves +0.051 AUROC—compared to +0.006 for the best delta configuration and negative results for the shared latent translator, establishing retrieval as the only architecture that produces clinically meaningful improvement on this task. On AKI, retrieval (+0.056 AUROC, +0.161 AUCPR) surpasses both delta and SL, reaching an absolute AUROC of 91.14 and AUCPR of 72.9—above even the in-domain MIMIC-native LSTM (89.7 AUROC, 66.5 AUCPR) van de Water et al. [2024]. The AUCPR improvement (+16.1 points) is nearly 3× larger than the AUROC gain (+5.6), indicating that the translator disproportionately improves precision at high recall—the regime most relevant for clinical deployment where false alarms carry real cost.

FeatureGate. We introduce a *FeatureGate* module, applicable across all three paradigms: learnable per-feature sigmoid weights $\sigma(w_f) \in [0,1]$ that modulate the reconstruction loss:

$$\mathcal{L}^{\text{gated}}_{\text{recon}} = \sum_{f=1}^{F} \sigma(w_f) \cdot \left(x^f_{\text{out}} - x^f_{\text{target}}\right)^2. \tag{9}$$

This automatically down-weights reconstruction error on features irrelevant to the downstream task (e.g., features the frozen model ignores), focusing capacity on task-critical dimensions. FeatureGate provides consistent gains: on mortality, SL+FG achieves +0.048 AUROC (+0.055 AUCPR) vs. plain SL +0.044 AUROC (+0.048 AUCPR).

4.4 Training Procedure. Training follows a two-phase protocol common to SL and retrieval translators. **Phase 1** (autoencoder pretrain, 15 epochs on MIMIC): the encoder–decoder is trained with reconstruction and label prediction losses. For retrieval, we introduce *self-retrieval pretraining*: during Phase 1, a memory bank is built from the encoder's own MIMIC representations

Table 3: Systematic comparison of domain adaptation and frozen-model methods across key desiderata for clinical time-series translation. **Model Frozen**: whether the target predictor's weights are entirely fixed (✓), partially fixed (P), or trainable (✗). **Adapt. Space**: Input (I), Representation (R), Weight (W), or Output (O). **Sample-Cond.**: whether the transformation depends on the input instance. **Clinical Tasks**: number of distinct clinical prediction tasks evaluated.

Method	Model Frozen?	Adapt. Space	Sample-Cond.?	Retrieval?	Var.-Len. TS?	Clinical Tasks	Grad. Theory?
Clinical DA							
VRADA Purushotham et al. [2017]	✗	R	✓	✗	✓	1	✗
OTTEHR OTTEHR Authors [2025]	✗	R	✗	✗	✗	3	✗
Mutnuri et al. Mutnuri et al. [2024]	✗	W	✓	✗	✓	3	✗
General TS-DA							
CoDATS Wilson et al. [2020]	✗	R	✓	✗	✗	—	✗
RAINCOAT He et al. [2023]	✗	R	✓	✗	✗	—	✗
CLUDA Ozyurt et al. [2023]	✗	R	✓	✗	✗	—	✗
Frozen-model / Input-space							
VPT Jia et al. [2022]	✓	I	✗	✗	✗	0	✗
Time-LLM Jin et al. [2024]	✓	I	✗	✗	✗	0	✗
TATO Qiu et al. [2026]	✓	I	P	✗	✓	0	✗
Retrieval-augmented							
kNN-MT Khandelwal et al. [2021]	✓	O	✓	✓	✓	0	✗
TS-RAG Ning et al. [2025]	✗	R	✓	✓	✓	0	✗
Foundation model							
ICareFM Burger et al. [2025]	✗	W	✓	✗	✓	4	✗
Ours (Retrieval Translator)	✓	**I**	✓	✓	✓	**5**	✓

and refreshed periodically. This provides non-degenerate context to cross-attention blocks from the start of training; without self-retrieval, cross-attention receives zero tensors during Phase 1, collapsing to a trivial pass-through that wastes the pretraining phase for the cross-attention parameters. **Phase 2** (retrieval-guided translation, 50 epochs): the full loss is:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{recon}}\,\mathcal{L}_{\text{recon}} + \lambda_{\text{range}}\,\mathcal{L}_{\text{range}} + \lambda_{\text{align}}\,\text{MMD} + \lambda_{\text{target}}\,\mathcal{L}_{\text{task}}^{\text{MIMIC}} + \lambda_{\text{label}}\,\mathcal{L}_{\text{label_pred}}, \tag{10}$$

where $\mathcal{L}_{\text{task}}^{\text{MIMIC}}$ applies the same task loss to MIMIC data through the frozen predictor, preventing representational drift. The memory bank is rebuilt every 3 epochs; learning rate follows cosine annealing Loshchilov and Hutter [2017]; gradient norms are clipped to 1.0; early stopping with patience 15 monitors the primary metric.

All data follows the YAIB preprocessing pipeline van de Water et al. [2024]: standardized variable selection (48 physiological features common to both databases), forward-fill imputation with missing indicators, per-feature normalization, and consistent patient-level train/validation/test splits. This ensures that distributional differences between eICU and MIMIC-IV reflect institutional variation rather than preprocessing artifacts. On top of YAIB's per-database normalization, we apply *cross-domain normalisation*: an affine transform that matches source per-feature mean and variance to target statistics prior to translation. This is computed once from training data and stored in the checkpoint, ensuring consistent normalisation at inference.

Phase 1 checkpoints are reusable: because pretraining depends only on the target data and model architecture (not Phase 2 hyperparameters), a single pretrain checkpoint serves all downstream experiments for a given task. We verify this empirically: across 9 AKI configurations varying K, W, L, and loss weights, the Phase 1 label prediction loss converges to 0.462–0.466, confirming Phase 2 independence.

4 Comparative Analysis and Contributions

With the related work surveyed in Section 1, we now position our work against prior art. We first provide a structured comparison across key desiderata (Section 4.1), then present focused head-to-head analyses with each method category (Section 4.2), situate our empirical results against reference baselines (Section 4.3), enumerate our concrete contributions (Section 4.4), and conclude with an honest assessment of the current scope and limitations (Section 4.5).

4.1 Systematic Comparison

Table 3 compares representative methods across eight dimensions that collectively define the desiderata for clinical time-series domain adaptation with a frozen predictor. No prior method satisfies all eight.

Several observations stand out. Representation-space (R) and weight-space (W) methods are architecturally coupled to the model's internals—they cannot be formulated without access to intermediate representations or weight tensors, violating the frozen-model constraint. In contrast, input-space translation treats the target model as a *black-box oracle* $f : \mathcal{X} \to \mathcal{Y}$. While

our efficient implementation leverages differentiable access (backpropagation through the frozen model) for gradient-based optimization, the method requires only query access to the model's input–output mapping. In principle, this makes input-space translation applicable even to proprietary models exposed only via API, using gradient-free optimization (e.g., evolutionary strategies or Bayesian optimization)—a property no R-space or W-space method can satisfy, though we note that gradient-free training of the translator remains untested and would likely require substantially more queries. The frozen-model methods from NLP and vision (VPT, Time-LLM) also operate in input space but use input-independent perturbations, limiting their capacity for heterogeneous clinical data. kNN-MT Khandelwal et al. [2021] is closest in spirit—frozen model, retrieval, per-timestep conditioning—but operates in output space (interpolating token distributions) rather than input space (transforming the data itself). Our retrieval translator is the only method that combines a fully frozen predictor, input-space adaptation, sample-conditional neural translation, target-domain retrieval, variable-length time-series support, evaluation across five clinical tasks, and a theoretical gradient alignment framework.

4.2 Head-to-Head Analysis

Against clinical DA methods. VRADA Purushotham et al. [2017] pioneered adversarial alignment for clinical time series but requires end-to-end training and was evaluated on a single task (mortality). Mutnuri et al. Mutnuri et al. [2024] address three of our five tasks (mortality, AKI, LoS) with inductive transfer and standard adversarial DA, but fine-tune the target model—precisely the operation our setting forbids. On the shared AKI task, our frozen-model retrieval translator achieves AUROC 91.14 and AUCPR 72.9, surpassing the MIMIC-native LSTM baseline (89.7 AUROC, 66.5 AUCPR) that represents the performance ceiling when the model is trained on in-domain data—despite never modifying any predictor weights. OTTEHR OTTEHR Authors [2025] applies optimal transport to the same eICU–MIMIC databases but performs population-level feature alignment without temporal modeling, treating each feature independently. Our instance-level, per-timestep retrieval and causal attention preserve the temporal structure essential for sequential clinical tasks (AKI, sepsis, LoS). In summary, prior clinical DA methods either sacrifice model integrity (fine-tuning) or temporal structure (population-level alignment); we sacrifice neither.

Against adapted DA baselines (DANN/CORAL/CoDATS). A natural question is whether standard DA methods can be adapted to our frozen-model setting by learning an input transformation $T_\theta(\mathbf{x})$ that precedes the frozen predictor, with DANN's gradient reversal Ganin et al. [2016] or CORAL's covariance alignment Sun and Saenko [2016] applied to the predictor's internal representations. We implement these adapted baselines (detailed in the full paper) and find that they achieve limited improvement. The structural argument is that clinical domain shift is *heterogeneous*: across 48 physiological features, the distributional shifts differ in magnitude, direction, and functional form (e.g., heart rate may require a simple affine correction while lab values exhibit nonlinear, context-dependent shifts). A shallow input transform with a single domain discriminator or covariance penalty lacks the capacity for such feature-specific, sample-conditional mappings. Our translator architectures provide structured inductive biases—additive residuals (delta), latent bottleneck (shared latent), or target-domain cross-attention (retrieval)—that decompose this complex mapping into learnable components. The gradient alignment analysis (Section 2) further explains the failure mode: when $\cos(\nabla_\theta \mathcal{L}_{\text{task}}, \nabla_\theta \mathcal{L}_{\text{fid}}) < 0$, as observed for sparse-label tasks, the domain-alignment and task objectives pull the input transform in opposing directions, a conflict that generic input transforms cannot resolve.

Against frozen-model methods. Prompt tuning Lester et al. [2021] and VPT Jia et al. [2022] learn a fixed set of $\mathcal{O}(1)$ parameters (soft tokens or visual prompts) that are shared across all inputs. Clinical EHR time series vary from 1 to 169 timesteps, span 48–292 features with complex missingness patterns, and exhibit patient-specific distributional characteristics—far exceeding the capacity of input-independent perturbations. Time-LLM Jin et al. [2024] reprograms frozen LLMs for time-series forecasting, but performs cross-*modality* adaptation (language→time series) for a generic pre-trained model, not cross-*domain* adaptation for a task-specific validated clinical predictor. TATO Qiu et al. [2026], a concurrent ICLR 2026 contribution, is the closest work in spirit: it explicitly proposes "adapt data to model" for frozen time-series foundation models. However, TATO relies on handcrafted transformation operators (slicing, normalization, outlier correction) rather than learned neural translation, targets forecasting rather than clinical prediction, and provides no theoretical analysis of when data adaptation succeeds or fails. We view TATO as validating the broader "adapt data, not model" paradigm; our work extends it with (i) learned end-to-end translation, (ii) retrieval-augmented cross-attention, and (iii) gradient alignment theory.

Against retrieval-augmented methods. kNN-MT Khandelwal et al. [2021] is our closest architectural ancestor: both systems pair a frozen model with per-timestep k-NN retrieval from a domain-specific datastore. The critical distinction lies in *how* retrieval is used. kNN-MT interpolates *output token distributions*—blending the frozen decoder's next-token prediction with the empirical distribution from retrieved neighbors. Our retrieval translator uses retrieved target-domain embeddings as *cross-attention context* in a Transformer decoder that transforms the *input*, enabling selective feature-level borrowing rather than output-level interpolation. Moreover, kNN-MT operates within a single language (e.g., domain-adapted German→English), while we bridge fundamentally different data distributions with heterogeneous recording practices and device calibrations. TS-RAG Ning et al. [2025] and RATD Liu et al. [2024b] retrieve time-series segments for forecasting augmentation within the same domain; our retrieval bridges *across domains*—a qualitatively different use of the retrieval paradigm. RAM-EHR Xu et al. [2024] retrieves textual medical knowledge from PubMed and knowledge graphs; we retrieve encoded *time-series windows* representing what the frozen predictor expects to see, grounding the translation in the target domain's statistical structure rather than external knowledge.

4.3 Results Positioning

Table 4 situates our best results against reference baselines from the YAIB benchmark van de Water et al. [2024]. Full experimental details and ablations will appear in the full paper.

Table 4: Our best translator results vs. reference baselines from YAIB van de Water et al. [2024]. Classification metrics: AUROC and AUCPR $\times 100$; regression: MAE in natural units. **Bold** indicates our result matches or surpasses the reference. †: MIMIC-native LSTM surpassed. All our results use a *frozen* MIMIC-IV predictor with no weight updates.

	Mortality		**AKI**		**Sepsis**		**LoS**	**KF**
	AUROC	AUCPR	AUROC	AUCPR	AUROC	AUCPR	MAE (h)	MAE (mg/dL)
Frozen baseline	80.79	50.2	85.58	56.8	71.59	3.0	42.5	0.403
eICU-native LSTM	85.5	35.7	90.2	69.9	74.0	4.0	39.2	0.28
MIMIC-native LSTM	86.7	41.0	89.7	66.5	82.0	8.0	40.6	0.28
Ours (best)	**85.55**	**55.7**	**91.14**†	**72.9**†	**76.71**	**5.2**	**39.2**	0.382
Δ vs. frozen	+4.76	+5.5	+5.56	+16.1	+5.12	+2.2	−3.3	−0.021

Where we exceed expectations. On AKI, our frozen-model translator achieves AUROC 91.14, surpassing not only the eICU-native LSTM (90.2) but also the *MIMIC-native LSTM* (89.7)—the model trained on in-domain data with full weight optimization. The AUCPR results are even more striking: 72.9 vs. 66.5 for the MIMIC-native LSTM and 69.9 for the eICU-native LSTM, a $+16.1$ point improvement over the frozen baseline. AUCPR is particularly informative for clinical tasks with class imbalance because it measures the precision–recall tradeoff directly: a higher AUCPR means the model identifies more true positives without generating excessive false alarms. The fact that the AUCPR gain ($+16.1$) is nearly $3\times$ the AUROC gain ($+5.6$) indicates that translation disproportionately improves the model's ability to distinguish positive cases at clinically relevant operating points. On mortality, translation pushes AUCPR to 55.7—far above both the eICU-native LSTM (35.7) and the MIMIC-native LSTM (41.0). Notably, the frozen baseline already exceeds both native models on mortality AUCPR (50.2), and the translator extends this advantage further. This demonstrates that input-space translation can recover performance *beyond* what the target model achieves on its own training distribution, suggesting that the translated source data may introduce beneficial distributional diversity that complements the frozen predictor's training distribution. All three classification tasks surpass the eICU-native LSTM on both AUROC and AUCPR, confirming that the translator closes the domain gap rather than merely compensating for it. On LoS (regression), the translator achieves MAE of 39.2 hours, matching the eICU-native LSTM exactly and improving upon the MIMIC-native LSTM (40.6 hours). Cross-server reproducibility is strong: across 3 seeds on 2 physically distinct GPU servers, AKI AUROC varies by only ± 0.0005, confirming that the gains are robust and not artifacts of hardware-specific numerical behavior.

Where we acknowledge gaps. We identify three areas where performance does not yet match the strongest reference points, and in each case, we characterize the underlying cause. *Sepsis*: the retrieval translator achieves $+5.12$ AUROC and $+2.2$ AUCPR, surpassing the eICU-native LSTM on both metrics (76.71 vs. 74.0 AUROC; 5.2 vs. 4.0 AUCPR—a 30% relative AUCPR improvement), but the gains are smaller in magnitude than on denser tasks. With only 1.13% positive labels per timestep, the task gradient is extremely sparse: our gradient alignment analysis ($\alpha = -0.21$ for sepsis vs. $+0.84$ for mortality) correctly predicts the relative difficulty and identifies label sparsity as the primary bottleneck for further improvement. *Kidney Function*: the KF task involves 292 features including 192 generated cumulative statistics (`_min_hist`, `_max_hist`, `_mean_hist`, `_count`) that must maintain internal consistency after translation—a substantially harder translation target. Our MAE of 0.382 improves over the frozen baseline (0.403) but a gap to the eICU-native reference (0.28) remains. *Single domain pair*: our current evaluation covers eICU→MIMIC-IV; extension to HiRID→MIMIC-IV is ongoing and will test generalization of the approach across a third institutional distribution.

4.4 Contributions

Our work makes five concrete contributions:

1. **Frozen-model preservation as a first-class constraint.** We formalize the setting where the *entire* target predictor is frozen—not just the classifier head (SHOT Liang et al. [2020]) or a subset of layers—and show that meaningful domain adaptation is achievable purely through input-space translation. This is directly motivated by the FDA's Software as a Medical Device (SaMD) framework U.S. Food and Drug Administration [2021a], which provides a simpler regulatory pathway for "locked" algorithms whose weights are not modified post-deployment.
2. **Input-space modularity.** The translator is an external, independently trainable module that can be swapped, updated, or audited without touching the validated predictor. This decoupling enables a deployment model where the predictor and translator have separate validation lifecycles—a practical requirement in regulated clinical environments.
3. **Instance-level retrieval-augmented cross-attention for clinical time-series DA.** To the best of our knowledge, we are the first to combine per-timestep k-NN retrieval from a target-domain memory bank with cross-attention decoding for domain adaptation of clinical time series. This provides each translated timestep with contextual grounding in what the frozen model expects to see, enabling the fine-grained, feature-specific transformations that generic input transforms cannot achieve.
4. **Universal paradigm across five clinical tasks.** We evaluate on the broadest task set in the clinical DA literature: three classification tasks (Mortality24, AKI, Sepsis) spanning per-stay and per-timestep label structures, and two regression tasks (Length of Stay, Kidney Function) with continuous targets. The retrieval translator achieves the best or tied-best results on all five, with improvements on both AUROC and AUCPR for all classification tasks—including surpassing the MIMIC-native LSTM on both metrics for AKI. This establishes retrieval translation as a universal paradigm with task-adapted capacity (specifically, the number of cross-attention layers: 3 for dense-label tasks like AKI, 2 for sparse-label tasks like sepsis).

5. **Gradient alignment as a predictive diagnostic.** We introduce the cosine similarity $\cos(\nabla_\theta \mathcal{L}_{\text{task}}, \nabla_\theta \mathcal{L}_{\text{fid}})$ as a diagnostic for frozen-model translation, connecting to the gradient conflict literature in multi-task learning Yu et al. [2020], Liu et al. [2021] and UDA Du et al. [2024], Phan et al. [2024]. This metric correctly predicts method–task suitability a priori ($+0.84$ for mortality, where delta translation succeeds; -0.21 for sepsis, where it fails and retrieval is required) and reveals that the retrieval architecture resolves the conflict *structurally* through the cross-attention information pathway, rather than requiring algorithmic gradient surgery.

4.5 Scope, Applicability, and Limitations

When to use input-space translation. Our approach is most applicable when three conditions hold: (i) a validated target-domain predictor exists and must remain frozen (regulatory, institutional, or practical constraints); (ii) labeled data from both source and target domains is available for translator training; and (iii) the target task provides sufficient label signal for the task gradient to guide translation. In our experiments, translation gains scale with label density: AKI (11.95%) and mortality (5.5%) yield the largest improvements, while sepsis (1.13%) shows smaller but still meaningful gains ($+5.12$ AUROC), consistent with the gradient alignment analysis. Under these conditions, the retrieval translator consistently closes the domain gap across diverse clinical prediction tasks.

When alternatives are preferable. If the target model can be retrained, standard domain adaptation or foundation model fine-tuning can optimize the predictor's own parameters—a strictly more expressive search space—and may outperform input-space translation, though this is not guaranteed: our translator surpasses the in-domain MIMIC-native LSTM on AKI, suggesting that the inductive bias of retrieval-augmented translation can sometimes compensate for the constrained parameter space. If no target-domain data is available, source-free methods (e.g., MAPU Ragab et al. [2023b]) are one option; alternatively, our delta translator can operate without a target-domain memory bank, using only the frozen model's task gradient and fidelity regularisation to learn input corrections—achieving up to $+2.9$ AUROC on AKI (vs. $+5.6$ with retrieval) when gradient alignment is positive ($\alpha > 0$). If real-time inference latency is critical, the memory bank k-NN lookup introduces approximately 2–3$\times$ overhead compared to the frozen model alone, which may be unacceptable in time-sensitive clinical alerting systems.

Current limitations. We identify five limitations that define the scope of the present work and inform future directions:

1. *Single domain pair.* Our evaluation covers eICU→MIMIC-IV. Extension to additional source domains (HiRID→MIMIC-IV) is under way and will test whether the approach generalizes across institutional distributions with different data collection protocols.
2. *Label density dependence.* Translation quality correlates with label density: AKI (11.95% positive rate) shows the largest gains (ΔAUROC $+5.56$, ΔAUCPR $+16.1$), while sepsis (1.13%) shows smaller but still meaningful improvements ($+5.12$ AUROC, $+2.2$ AUCPR). The gradient alignment diagnostic quantifies this relationship, but does not resolve the underlying information bottleneck for extremely sparse tasks.
3. *Calibration degradation.* Translation improves discrimination (AUROC, AUCPR) but can degrade probability calibration (ECE), requiring post-hoc temperature scaling—an additional validation step in clinical deployment.
4. *Inference overhead.* The retrieval translator requires maintaining a GPU-resident memory bank and performing k-NN search at each timestep, adding computational and memory costs relative to direct model inference.
5. *Architecture specificity.* Our frozen baselines are LSTMs from the YAIB framework van de Water et al. [2024]. Since input-space translation treats the predictor as a black-box function $f : \mathcal{X} \to \mathcal{Y}$, the paradigm is architecture-agnostic in principle; empirical validation with other target architectures (Transformers, GRUs, temporal convolutional networks) remains future work.

We view these as open problems rather than fundamental obstacles. The frozen-model, input-space translation paradigm opens a new direction in clinical DA research, and the gradient alignment framework provides guidance for extending it to new domains, tasks, and architectures.

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